

Classifiers and Plurality in Languages of the Americas*

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1. Introduction

In this paper, we discuss the co-existence of classifier and plural marking in three languages of the Americas. By classifiers, we refer broadly to morphemes that appear in the presence of numerals, and which, in many languages, track physical properties of the objects being counted, such as the following examples from Ch'ol, a Mayan language of Mexico, and Máíhĩkì, a Tukanoan language of Peru.¹

(1) Examples of classifiers

- a. Ta' paj-a [cha'-**k'ej** waj].
PFV sour-INCH two-CL:flat tortilla

'Two tortillas became moldy.'

Ch'ol

- b. [tèpè -**tara**] chíá -bi
two -CL:rigid.cylinder buy -1PL.PAST.DECL

'We bought two bottles.'

Máíhĩkì (Farmer 2015:156-7)

The role that classifiers play in the noun phrase has long been of interest to work in formal semantics (see, for example, Krifka 1989, 1995, Chierchia 1998a, 1998b, among many others). Early investigations into classifiers focused primarily on languages of Asia,

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¹Glossing conventions: 1 = first person; 2 = second person; 3 = third person; A = set A markers in Mayan languages index possessive and ergative arguments; ACC = accusative; AGT = agentivizer; AI = animate intransitive stem in Algonquian languages; AN = animate; B = set B markers in Mayan languages index absolutive arguments; CL = classifier; DECL = declarative; DEM = demonstrative; DIF = different; EMPH = emphatic particle; II = intransitive inanimate stem in Algonquian languages; INAN = inanimate; INCH = inchoative; INTR = intransitive; IV = intransitive verb (Mayan); MASC = masculine; NI = *-ni* class inflectional paradigm in Máíhĩkì; PERF = perfect; PFV = perfective aspect; PL = plural; PLACT = pluractional; PREP = preposition; PRES = present; PRT = particle; PURP = purposive; SP: = Spanish-borrowed word; SUBJ = subject; TI = transitive inanimate stem in Algonquian languages

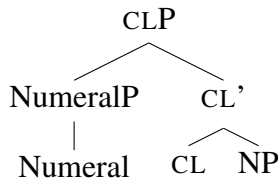
tying their distribution to countability distinctions in nouns and inspiring a variety of analyses that make clear predictions about the interaction of classifiers and plurality. Some of these analyses predict that classifiers and plural marking do not co-occur in a single noun phrase, accounting for their complementary distribution in the presence of numerals in some languages (Chierchia 1998b,a, Borer 2005).

Recent investigations into a wider range of languages have revealed a more complex empirical picture in the behavior of both classifiers and plurality. For instance, Little et al. (2022) argue that classifiers are not a homogeneous grammatical category. That is, they argue that some classifiers are for nouns and some classifiers are for numerals, as shown in the trees in (2).

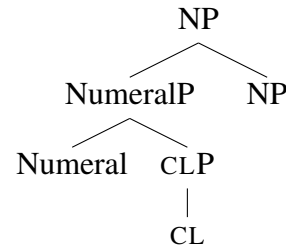
(2) Two types of classifiers

(based on Little et al. (2022))

a. Classifier-for-noun



b. Classifier-for-numeral



In analyses such as those in Chierchia (1998b) and Borer (2005), the structure of numerals and classifiers is like that in (2a), where classifiers modify the meaning of the noun, allowing it to compose with a numeral. On Borer's analysis in particular, plural marking is in complementary distribution with classifiers as they both are housed in the same head along the nominal spine and perform the same semantic role: they individuate noun meanings. For classifier-for-numeral languages, there is no such prediction on the complementarity of classifier and plural marking: nothing about the structure in (2b) precludes plural marking from appearing as classifiers are selected by numerals.

In this paper, we discuss three languages of the Americas—Ch'ol (Mayan, Mexico), Mi'gmaq (Algonquian, Canada) and Máihĩkì (Tukanoan, Peru)—in light of their classifier and plural marking systems. Crucially, all three languages allow the co-occurrence of classifiers and plural marking. While Ch'ol and Mi'gmaq have been discussed before in the theoretical literature on classifiers (Bale and Coon 2014, Bale et al. 2019, Little et al. 2022), as prime examples of classifiers-for-numerals languages, Máihĩkì has not received much attention. For Ch'ol and Mi'gmaq, we contextualize our discussion not just on classifiers, as previous works have, but also on the existence of plural marking, adding in new data from Ch'ol on plurals. In Máihĩkì, we present a different type of classifier system from those of Ch'ol and Mi'gmaq—one in which classifiers modify nouns—but see again that classifiers and plurals co-occur. Together, the co-occurrence of classifiers and plurality in these three different systems suggest an even more complex empirical landscape than previously thought.

This paper is structured as follows. First, we discuss background literature on classifiers and plural marking and theories which predict a complementarity relationship between

them (Section 2). We then discuss Mi'gmaq and Ch'ol, where classifiers can co-occur with plural marking, but in different ways (Sections 3.1 and 3.2). These do not pose a direct counterexample to claims on complementarity of classifiers and plural marking if we adopt a particular analysis of classifiers. We then discuss Máhĩkì, which does seem to pose more challenges for a Borer-style analysis of classifiers and plural marking (Section 3.3). We end the paper in Section 4 with a discussion of the three languages and possible paths forward for the study of classifiers and plural marking.

2. Background

In this section, we present background on classifiers and plural marking that led researchers to propose analyses that predict their complementary distribution. We focus on works such as Chierchia (1998a,b) and Borer (2005) who make predictions about the complementary distribution of classifiers and plural marking.

2.1 The interaction of plurality and classifiers

One factor that has been central to theorizing the role of classifiers is the status of plural marking in these languages. Notably, many languages with classifiers lack the sort of plural marking that non-classifier languages like English have. For example, nouns in Mandarin Chinese—a prototypical classifier language, as cited in the literature—are number neutral; plural marking is not required in plural contexts.² For example, the noun *haizi* ‘child’ in sentence (3) bears no plural marking, but is number neutral in its interpretation.

- (3) wo qu zhao **haizi**
I go find child
‘I will go find the/some child/children.’ (Li 1999:78)

In other languages, classifiers and plural marking are both found, but cannot appear in a single noun phrase. In Western Armenian, for instance, noun phrases may contain plural marking, or classifiers, but never both (Borer 2005, Bale and Khanjian 2008). Example (4a) shows that Western Armenian has plural marking, while example (4b) shows that it also has classifiers (albeit optional ones). Example (4c) shows that classifiers and plural marking cannot appear in the same noun phrase.

- (4) Western Armenian (Borer 2005:94-95)
- a. Yergu havanoc-**ner** uni-m.
two umbrella-PL have-1SG
‘I have two umbrellas.’
 - b. Yergu **had** havanoc uni-m.
two CL umbrella have-1SG
‘I have two umbrellas.’

²Mandarin does have a plural marker *-men*, but it differs significantly its meaning and distribution from plural marking in languages like English (Li 1999, Kim and Meng 2022, among others).

- c. * Yergu **had** havanoc-**ner** uni-m.
 two CL umbrella-PL have-1SG
 ‘I have two umbrellas.’

Deriving the apparent complementary distribution between classifiers and plural marking has been a central goal of a variety of theories about the role of classifiers in the noun phrase. For example, Borer (2005) proposes that plural marking and classifiers perform a similar semantic role, namely, one of individuation. On her analysis, the two never appear in the same noun phrase because they instantiate the same functional head in the DP: the one associated with division of mass meanings into countable units.

While the generalization that classifiers and plural marking are in complementary distribution holds true for specific languages, several counterexamples have been identified in the literature more recently, including Persian (Gebhardt 2008) and Indonesian (Dalrymple and Mofu 2012), both of which have optional classifiers, and Tiwa (Dawson 2020), which has obligatory classifiers. All three languages allow plural marking and classifiers within a single noun phrase. This is exemplified for Tiwa in (5), which shows the plural marker *-râw* appearing in the same noun phrase as the general count classifier *-tha*.

- (5) Maria [thin-**tha** lái-**râw**-go] pre-ga.
 Maria three-CL book-PL-ACC buy-PFV
 ‘Maria bought three books.’ (Dawson 2020:48)

The contrast between languages like Western Armenian and languages like Tiwa suggests that the empirical picture surrounding how classifiers and plurals interact is more complicated than previously thought.

2.2 The heterogeneity of classifiers and plural marking

It is complicated to assess the generalization that Borer’s analysis rests on. As the typological literature has long noted (e.g., Corbett 2000, Aikhenvald 2000, Gil 2013), neither classifiers nor plural marking are homogeneous classes across languages, and this heterogeneity has been recognized more recently in the formal literature. In this section, we will discuss the key ways in which classifiers and plural marking differ across languages. We also note how these differences complicate conducting large-scale typological surveys on the distribution of classifiers and plurality based on the level of description available for most languages.

The term ‘classifier’ is used in linguistics to refer to a broad range of morphemes that track various properties of the head noun. In the typological literature, it has been shown that there are different types of classifiers, including numeral classifiers and noun classifiers (see, e.g., Aikhenvald 2000). The formal semantic literature referenced above is primarily concerned with analyzing numeral classifiers, which are required in a noun phrase when the noun is modified by a numeral. Within this formal literature, there are two broad views on the function of numeral classifiers: (i) that classifiers modify noun meanings, for example, by individuating them (and thus allowing them to be modified by numerals; Chierchia 1998a, Borer 2005), or (ii) that classifiers modify numeral meanings in such a way that

The degree of heterogeneity in classifiers and plural marking makes large-scale typological surveys that explore how they interact difficult. The descriptions available for most languages of the Americas that we have encountered do not provide enough information to assess what kind of classifiers or plural marking the language has. Moreover, a language might be easily miscategorized due to morphosyntactic differences could be missed. For example, languages like Guató (isolate, Brazil) may not have any markers of plurality that appear within the noun phrase and might therefore be taken to be a language that lacks plural marking. However, Guató has verb agreement that tracks the plurality of third person subjects (Godoy and Alves 2020), suggesting that noun phrases in the language do bear number features just as they do in languages like English. An example from Guató is provided below. The bolded prefix *bε-* indexes the third person plural subject of a transitive verb.

- The formal semantic literature that explores the interaction of classifiers and plural, discussed in section 2.1 above, does not fully take into account the complications posed by this more complex empirical landscape. This leaves us with the question: does the generalization that classifiers and plural are in complementary distribution apply to particular kinds of classifiers and plural marking? While large scale typological surveys are difficult, more in-depth studies of classifiers and plurality in languages of Americas reveal that classifiers and plural marking of different sorts can co-occur. Therefore, yes, the empirical picture of classifiers and plural marking is complex and while plural marking and classifiers may be in complementary distribution in one language, they may co-occur in other languages. We expound on this with examples from three languages where plural marking and classifiers may co-occur.

³See Dékány (2024) for some complications regarding the diagnostics in Little et al. (2022).

3. Three case studies

In this section, we discuss three cases of languages of the Americas: Mi'gmaq, Ch'ol, and Máhĩkì. These three languages have very different kinds of classifier systems. They also all have plural markers with different morphosyntactic (and possibly semantic) profiles. Despite these differences, all three seem to be counterexamples to the Chierchia-Borer generalization: classifiers and plural marking can co-occur within a single DP in each language.

3.1 Mi'gmaq

Mi'gmaq is an Algonquian language spoken in Eastern Canada by about 7,000 people.⁴ Bale and Coon (2014) describe it as a classifier language, though only some numerals require classifiers. Numerals 1–5 do not require the classifier *te's*, shown below in (7) for numerals two and three; numerals above six require the classifier *te's* as shown in (8) for the numeral six. The examples were taken from the Mi'kmaq Online Dictionary (Haberlin et al. 1997–2025). The italicized text to the right of the translation indicates the entry in Mi'gmaq under which the example was found. Mi'gmaq nouns are either grammatically animate or inanimate, the examples below show that the pattern is consistent across grammatical animacy categories⁵: an example with an animate and inanimate noun is shown for each type of numeral. The examples below indicate that obligatory plural marking co-occurs with the numerals. The numerals appear with additional morphology such as the intransitive suffixes for animate intransitives and inanimate intransitives, third person markers and plural markers.

(7) Numerals without a classifier in Mi'gmaq (1–5 and numerals morphologically built from them)

- a. [Tapus-i-j-ig lapol-g] ep-i-tj-ig.
 two-AI-3-PL.AN bowl-PL.AN lie-AI/II-3-PL.AN
 'There are two bowls there.' *lapol* 'bowl'
- b. Lpa temina'-to-q wi'gatign [ne'sis-g-l mtijin-n] telpase'-g.
 EMPH rip-TI-3 book three.II-3-PL.IN inch-PL.IN deep.II-3
 'He/she ripped a book three inches thick.' *temina'toq* 'rip'

(8) Numerals with the classifier *te's* (6 and higher)

- a. [As'gom **te's'-g-l** sigsuti'-l] wap-e'-g-l
 six CL-II-3-PL.IN ankle.length.sock-PL.IN white-AI/II-3-PL.IN
 pgwatele'n.
 buy.IMP
 'Buy six white ankle length socks.' *sigsuti* 'ankle length socks'

⁴According to statistics published in 2016 by Statistics Canada. <https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/hlt-fst/lang/Table.cfm?Lang=E&T=41&Geo=01>

⁵While we do not show semantically animate examples, which are grammatically animate as well, we note that they, too, obligatorily mark plural.

- b. [As'gom **te's**-i-j-ig tmato's-g] tms.
 six CL-AI-3-PL.AN tomato-PL.AN cut.IMP
 'Cut six tomatoes.' *tmatos* 'tomato'

The data in Mi'gmaq are significant for the study of classifiers and plural marking because they display evidence for a language that has classifiers, at least for some numerals, and the co-existence of *obligatory* plural morphemes. As we discuss below for Ch'ol, if classifiers are analyzed as dependent on the numeral—that is, certain numerals need classifiers, rather than nouns—then this may not be a counterexample to the Chierchia-Borer hypothesis. We discuss this more in section 3.2.

Finally, the morpheme *te's* analyzed as a classifier in Mi'gmaq by Bale and Coon (2014), may occur in the absence of numerals, such as in the examples below. In (9a) it appears quantity work to mean 'many'. In (9b) it appears with *ta'n*, a complementizer to form the quantifier 'everyone.'

- (9) *Te's* in the absence of numerals in Mi'gmaq
- a. Na **te's**-'g-l munti-l pew-atm-an-n.
 PRT CL-II-3-PL.IN bag-PL.IN want-TI-1-PL.IN
 'I want this many bags.' *te's*'g 'so much/so many'
- b. Ta'n **te's**-i-t wen teluisi-t Mali...
 COMP CL-AI-3 who is.called.AI-3 Mary
 'Everyone who is called Mary...' *te'sit* 'so much / so many'

Unlike what we will see below for Ch'ol and Máfhkì , the classifier *te's* does not vary depending on what is being counted.

3.2 Ch'ol

Ch'ol is a Mayan language of the Ch'olan-Tzeltalan subgrouping spoken in southern Mexico and diasporic communities in North America by 250,000 people.⁶ Ch'ol, like Mi'gmaq, also displays evidence for the co-occurrence of classifiers and plural marking, but plural marking is not obligatory in the same way it is in Mi'gmaq. As per Bale and Coon (2014), classifiers appear with Ch'ol-based numerals, but not Spanish borrowed ones. Vázquez Álvarez (2011:160) states that speakers of Ch'ol, even monolingual speakers, will use Ch'ol-based numerals 1–6, but above 6, speakers will employ Spanish-based numerals. Examples are shown below of numerals in Ch'ol. (10a) shows a Ch'ol-based numeral with the classifier for flat objects and (10b) shows a Spanish-based numeral with no classifiers. Classifiers only appear with numerals and the interrogative numeral *jay* 'how many'.⁷

(10) Numerals in Ch'ol

⁶These statistics are from Mexico's 2020 census: https://cuentame.inegi.org.mx/descubre/poblacion/hablantes_de_lengua_indigena/

⁷Uncited data below comes from author Little's fieldwork with Ch'ol between 2024 and 2025.

- a. Ta' paj-a [cha'-k'ej waj].
PFV sour-INCH two-CL:flat tortilla
'Two tortillas became moldy.'
- b. Juñyaj chañ [syete alob].
EMPH tall SP:seven boy
'Seven boys are quite tall.'

As suggested by the glosses in (10), plural marking is not required with numerals greater than one, even for human-denoting nouns. However, the plural suffix, *-ob* may occur, as shown in (11a) where it can appear on the predicate and/or the noun (see also Little et al. 2022:18–19). The numeral appears with the classifier for humans, exhibiting the co-occurrence of classifiers and plural marking. The example in (11b) is also possible, where there is no plural marking, suggesting that plural marking is optional.

(11) Ch'ol plural is optional with numerals above one

- a. Ta' jul-i-**yob** [cha'-tyikil x-cholel-**ob**].
PFV arrive-IV-PL two-CL:human AGT-field-PL
'Two farmers arrived.'
- b. Ta' jul-i [cha'-tyikil x-cholel].
PFV arrive-IV two-CL:human AGT-field
'Two farmers arrived.'

Vázquez Álvarez (2011) reports that the plural marker *-ob* is mostly used with human-denoting referents but has observed instances where it occurs with animals and insects. One of his examples is the naturally occurring one in (12), where *-ob* appears on the intransitive verb and indexes the plurality of the noun, grasshopper, which appeared earlier in the recording. There is an epenthetic *y* (IPA: [j]) on *-ob* to break up vowel hiatus.

- (12) ya' tyi kej-i-**yob** tyi tyuñ
there PFV start-IV-PL PREP lay.egg
'They (the grasshoppers) started to lay eggs there.' (Vázquez Álvarez 2011:85)

Vázquez Álvarez (2011) points out that there is dialectal, and even individual, variation on whether *-ob* can indicate non-human referents. In my fieldwork with a young Tumbalá speaker (20 years old), she accepted the following sentence, where plural may appear on the predicate, but notably not the noun, which is inanimate.

- (13) Ta' paj-a-**yob** aläxax(*-**ob**).
PFV sour-INCH-PL orange-PL
'The oranges became moldy.'

In order for a plural interpretation to be possible for human-denoting nouns in the absence of other quantity word appears, *-ob* must occur on the predicate or the noun. The example in (14) shows that when there is no *-ob* on a bare noun or the predicate, a plural reference is not possible. In (14) when speaker A introduces the bare noun *wiñik* 'man'

with no plural marking either on the predicate or noun phrase, speakers interpret this as singular. This is why, when A answers with the names of three people, it is infelicitous. This example suggests that plural is not always optional in Ch'ol.

(14) Conversation between A & B Ch'ol

a. A: Ta' k'oty-i wiñik tyi k'iñijel.
PFV arrive-IV man PREP party

A: 'A man arrived at the party.'

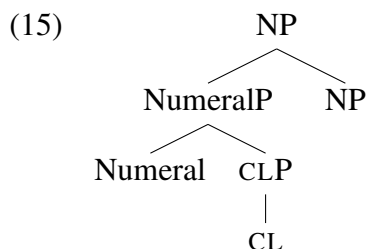
b. B: Majki?
who

B: 'Who?'

c. # Pedro, xWañ y-ik'oty xNico.
Pedro Juan A3-with Nico

A: 'Pedro, Juan and Nico'

While a detailed analysis of the Ch'ol plural morpheme is outside of the scope of this paper, one possibility for why the plural marker is needed in (14) but not in (10), where the numeral 'two' appears, is that in the absence of any plural marker or other plural quantity word, a singular inference arises, as per the analysis for optional plural in Tiwa in Dawson and Gibson (to appear). Effectively, the singular interpretation of (14) is an implicature. The data in Ch'ol support the fact that a classifier language may have plural and that that plural is obligatory in some contexts. Taken in hand with the analysis of Bale and Coon (2014) and Little et al. (2022) of Ch'ol where classifiers are needed to supply a measure function for the numeral, such as in the structure in (15), there is no prediction as to whether a language can have both classifiers and plural marking.



Under this analysis, it is not unsurprising that Mi'gmaq and Ch'ol can have both classifiers and numerals, thus not raising strong arguments against a Borer-style analysis. Plural may be housed within the extended nominal structure.

The pieces of data in this subsection suggest that Ch'ol plural is more complicated than simply being 'optional.' It is optional with numerals and other quantity words denoting pluralities, but preliminary data suggest it is obligatory in the absence of any other quantity words and may be an inflectional morpheme—either on the noun or the predicate—as it is needed in some contexts.

3.3 Máfhĩkì

Máfhĩkì is a Western Tukanoan language spoken by approximately 80-90 people in north-east Peru (Farmer 2015). The data and generalizations discussed here are based on Farmer's (2015) detailed description of its classifiers and plural marking. Máfhĩkì has an elaborate system of classifiers that differ significantly in their behavior from those of Mi'gmaq and Ch'ol: while classifiers in those languages appear to modify numerals, classifiers in Máfhĩkì modify nouns. This is evident from the fact that classifiers can appear with just a noun, as shown in (16), where the classifier *-ri* appears with the noun *kíù* 'metal'. Further, classifiers can and do modify noun meaning: they are used to derive count nouns from mass nouns, as in this example.

- (16) [*nómìò kíù -ri*] *tíyí -hú -gi*
 woman metal -CL:manufactured crack.INTR -PERF -3SG.MASC.PAST.DECL
 'The woman's machete cracked' (Farmer 2015:98)

Classifiers in Máfhĩkì track various properties of the noun, including gender for animate nouns and shape for inanimate nouns. Farmer (2015:132) lists 17 'full classifiers', which are only used as classifiers; Máfhĩkì additionally has a large number of classifiers which are homophonous with other roots (such as nouns and verbs).

In addition to appearing on mass nouns, classifiers in Máfhĩkì obligatorily appear on nominal modifiers, including demonstratives, numerals, and adjectives. All three of these categories, along with a classifier, can appear in argumental position without an overt noun. This is shown in (17), where the numeral 'two' along with a classifier serves as the object of the clause.

- (17) [*tèpè -tara*] *chíá -bi*
 two -CL:rigid.cylinder buy -1PL.PAST.DECL
 'We bought two bottles.' (Farmer 2015:156-7)

Note that while count nouns do not themselves appear with classifiers, their modifiers do.

In addition to its robust system of classifiers, Máfhĩkì also has plural marking. Plural in the noun phrase can be marked via two suffixes: *-ma* for inanimates, shown in (18), and *-na* for animates, shown in (19). There are also plural forms of some nouns and some classifiers.

- (18) [*wèè -ma*] *tátá -hea -gi*
 house -INAN.PL fall.PLACT -PERF.PLACT -3SG.MASC.PAST.DECL
 'The houses fell down' (Farmer 2015:177)

- (19) [*kòrè -na -re*] *títé -yi*
 spider.mite -AN.PL -NON.SUBJ extract-PLACT -1SG.PRES.DECL
 'I'm extracting spider mites.' (Farmer 2015:177)

Non-plural-marked nouns can receive number-neutral interpretations in certain contexts, such as when they are the subject of a stative predicate or an object in particular syntactic configurations. However, non-plural-marked nouns can only be interpreted as singular with

episodic verbs, and pluractional telic verbs require plural-marked subjects. Additionally, animate plural subjects trigger plural agreement on the verb (as in (21) below; inanimate plural subjects do not trigger plural agreement).

Crucially, classifiers and plural marking can co-occur in Máfhĩkì. This holds for both inanimates, as shown in (20), and animates, as shown in (21).

- (20) yíkí [mǐí -toto -ma] bèè -yi
 1PL palm.PL -CL:flat.rigid -INAN.PL carry.on.back -1PL.PRES.DECL
 ‘We are carrying the *cumbas*.’⁸ (Farmer 2015:177)

- (21) [ímì -na tèpè -aki -na] bàì nògì -yohì
 man -AN.PL two -CL:masc.being -AN.PL meat fish.with.hook -PL.PURP
 sá -hì yàrì -ya
 go.PAST.NI -3PL.PAST.DECL.NI small -CL:river
 ‘Two men went fishing in a stream’ (Farmer 2015:157)

While classifiers in Máfhĩkì behave differently than those in Asian languages (e.g., in marking modifiers as well as nouns), they clearly perform some sort of role in individuating noun denotations: they derive count nouns from mass nouns. This places Máfhĩkì classifiers squarely in the ‘classifiers-for-nouns’ category that Chierchia’s and Borer’s analyses assume. The data show that plural marking is available even in noun phrases that contain a classifier, presenting a true counterexample to the generalization that classifiers(-for-nouns) and plural marking are in complementary distribution.

One possible explanation for the co-occurrence of classifiers and plural marking in Máfhĩkì, that may maintain Borer’s generalization, could draw on Wiltschko’s (2008) proposal that plural marking in some languages is adjoined, rather than inflectional. On this analysis, plural marking would not occupy the same syntactic position as the classifier, which realizes a head associated with individuation, and presumably would perform some other semantic role. However, plural marking in Máfhĩkì cannot be plausibly analyzed as adjoined, according to Wiltschko’s own diagnostics. First, it is not optional in at least some syntactic environments. Second, it triggers plural agreement on the verb. Both behaviors are indicative of inflectional plural marking.

The co-occurrence of inflectional plural marking and noun classifiers demonstrates that they do not occupy the same morphosyntactic slot in Máfhĩkì. Indeed, classifiers seem lower in the structure than plural marking in Máfhĩkì, given the relative order of classifiers and plural suffixes. One possibility is that classifiers in Máfhĩkì instantiate a little *n* head. Such an analysis would explain their ability to nominalize verb roots, as shown in (22), where the classifier *-pi* nominalizes the verb root *tóyá* ‘write’. It would also provide a ready explanation for why Máfhĩkì’s classifier system shows, according to Farmer’s description and typological analysis, some hallmarks of a gender system: gender is very plausibly associated with *n* (Kramer 2016).

- (22) [mìnà tóyá -pi -ma] tóyá -yi
 DEM.DIF write -CL:stack -INAN.PL write -1SG.PRES.DECL

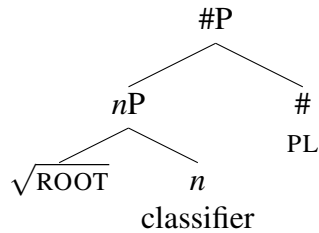
⁸Farmer (2015:177) notes that “a *cumba* is a cap of woven leaves placed on the peak of a roof.”

‘I’m writing in different notebooks.’

(Farmer 2015:50)

If this analysis is appropriate, we expect classifiers and plural marking to co-occur, instantiating *n* and a number projection respectively, as schematized in (23)

(23) Tentative analysis of classifiers and plurality in Máíhĩkì:



4. Conclusions

The case studies discussed above highlight the diversity of what is labeled a ‘classifier’ and what is labeled ‘plural’. Mi’gmaq has obligatory plurals for all nouns and classifiers appear with some numerals. Ch’ol has numeral classifiers and a plural that is optional in many instances but seems to be necessary in some contexts without any other quantity-denoting word that implies plurality. The Máíhĩkì data suggests that classifiers are distinct from those in Ch’ol and Mi’gmaq: they are more akin to classifiers-for-nouns as they can appear in environments even in the absence of any numeral. These languages display three very different ways in which classifiers and plurals co-occur.

If we take a classifier-for-numeral analysis for Ch’ol and Mi’gmaq, then it is not surprising that plural marking can co-occur, as they perform fundamentally different roles in the noun phrase. Thus, these languages do not present strong counterexamples for the Chierchia/Borer generalization as classifiers perform a different syntactic and semantic role. However, Máíhĩkì poses a greater challenge for the proposal that classifiers(-for-nouns) are in complementary distribution with plural marking, suggesting that even classifiers that do perform a role in individuating noun meanings can coexist alongside plural marking, even within the same noun phrase.

Lastly, it’s worth noting that much work on classifiers and plural marking has concentrated on east Asian languages. We would like to highlight that the empirical findings here suggest that generalizations from languages such as Mandarin are not as representative as the literature may lead a reader to believe. For instance, the fact that plural marking and classifiers *must* co-occur in some languages is evidence that classifiers and plural marking perform different functions than what has been reported in the literature for other languages. We hope to have highlighted the importance of a case-by-case studies of classifiers and plural marking. This is especially crucial for languages of the Americas, which are relatively under-represented in this literature.

We hope to have highlighted in this paper that classifiers and plural marking are not unified classes and careful semantic fieldwork on both these topics will continue to further our understanding of the diversity of classifiers and plural marking in languages of the world. In particular, any description of classifiers and plurality should attend to whether those elements are optional or obligatory and whether they can co-occur. Additionally, it is

important to understand the full distribution of classifiers in order to establish whether they form a constituent with the numeral or with the noun.

While we have hoped to highlight some of the complicating factors in assessing the status and interaction of classifiers and plural marking across languages, there is plenty more work to be done in understanding the full empirical landscape. In particular, all three languages we have examined here have obligatory classifiers, but many languages of the world have optional classifiers (e.g., Persian and Indonesian; Gebhardt 2008, Dalrymple and Mofu 2012). Similarly, animacy in some languages, including Ch'ol and Tiwa (Dawson and Gibson to appear), affects the distribution of plural marking. Further work is needed to explore how factors such as these change the broader typological picture.

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